

Multimodal Explainable AI (XAI) for Hateful Meme Classification: A Comprehensive Framework for Transparent Content Moderation

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Abstract—The high velocity of memes related to hate speech spread on the Internet is a major challenge to automated content control systems. Their destructive significance is frequently based on the intricate combination of the image and text, so these memes cannot be classified using the models of classifications, which examine each modality separately. Recent multimodal models have been found to achieve better classification accuracy; but are commonly used as “black boxes,” they are not transparent in their decision-making process. This paper presents a complete multimodal explainable AI (XAI) system that does not only classify memes as either hateful or non-hateful but also provides faithful, multimodal explanations of how the action came to this conclusion. The approach uses a Vision-and-Language Transformer (ViLT) model as well as novel explanatory mechanisms that take cross-modal attention patterns. An overall assessment of the Facebook Hateful Memes dataset indicates competitive predictive precision (accuracy = 0.742, AUC-ROC = 0.812) and quantitatively measures faithful explanations (Faithfulness = 0.234, AOPC = 0.156). The ablation test conducted with exhaustive analysis high maintaining the performance and explainability. The usefulness of the explanations is testified by human evaluation which involved 100 participants and showed high mean scores in plausibility (4.12/5.0) and helpfulness (4.28/5.0). This research, therefore, defines a confirmed route towards transparent, responsible and reliable AI systems in content moderation sphere.

Index Terms—Explainable AI, Multimodal Learning, Hateful Memes, Content Moderation, Transformers, Attention Mechanisms, Trustworthy AI

I. INTRODUCTION

The growth of social media platforms has turned internet memes from a cultural artifact to a powerful tool of hate speech, harassment, and misinformation. This growth has turned memes into important channels of hate speech and fake news. The multimodal nature of memes poses a special computational challenge: the hateful connotations are often not seen in the image or text alone, but in their specific, contextual combination [1]. An example is that a visual representation of a religious icon that is neutral is rendered hateful when it has derogatory text and the same text is not hateful in a different visual image.

The research community has reacted to this challenge by providing multimodal fusion architectures of enhanced sophistication. VisualBERT-based models [2], ViLT-based models [3], and CLIP-based models [4] have shown impressive results on the benchmarks, including the Facebook Hateful Memes dataset [1]. However, there is a price to such developments: as models continue to become more complex, interpretability is lost, leading to black-box systems that give classification results without other justification. This opaceness is especially an issue with regard to content moderation, in which the result of decisions has serious social implications and requires transparency to establish accountability, bias detection, and user trust [5].

Explainable AI (XAI) provides viable solutions to this issue, but the majority of current approaches have weaknesses in multimodal settings. Unimodal XAI methods like LIME [6] and SHAP [6] have trouble modeling cross modal interactions; attention based explanations in transformers [7] have been criticized for their faithfulness. The main issue it faces is to come up with explanations that not only bring out the salient features in each modality but also explain the interaction of such features to give the final classification.

This study solves the above mentioned challenges by three main contributions:

- 1) **An enhanced multimodal XAI framework** combines the ViLT architecture with advanced explanation mechanisms, using layer-wise attention aggregation and cross-modal relevance propagation, shown in the Figure 1.
- 2) **Comprehensive evaluation methodology** involving a mixture of quantitative measures of faithfulness, namely Faithfulness and AOPC, and a large scale human assessment that involves 100 respondents and thus involves a measure of both technical fidelity and usefulness.
- 3) **Extensive experimental analysis** involves ablation experiments, cross dataset testing, and a critical analysis of failure instances, which combined together offer future research undertakings.

The empirical results show that transparent multimodal

classification is not only possible for transparent classification, but increases performance and trustworthiness. The suggested system achieves better explainability scores yet maintains competitive classification scores, thus forming a proven base on deployable content moderation systems.

II. RELATED WORK

A. Multimodal Hateful Content Detection

The progression of hateful meme detection is part of general trends in multimodal learning. Early approaches used feature-level fusion strategies, which combine embeddings from pre-trained visual models (e.g. ResNet [8]) and language models (e.g. BERT [9]) using either concatenation or element-wise operations [10]. These approaches were computationally tractable, but they did not resolve the intricate interactions between modalities, which define hateful memes.

The Hateful Memes dataset [1] was a turning point where a benchmark has been provided which is designed specifically to challenge the unimodal classifiers with carefully constructed "confounder" examples. This catalysed the development of more sophisticated fusion architectures, in particular transformer based models, pre-trained on large scale vision language corpora. VisualBERT [2] and VL-BERT [11] handle region-based features of images along with text tokens using unified transformer encoders that help to perform deeper cross modal understanding. Later architectures like LXMERT [12] made use of distinct modality-specific encoders before going to layers of cross-modal fusion.

A huge improvement was brought forward by ViLT [3], which removed the computational burden of object detection by considering image patches as sequence tokens similar to text. Not only was efficiency enhanced in this patch-based method, but more verbose (granular) visual representation of data was provided, which was more likely to generate detailed explanations. More recently, large scale foundation models such as CLIP [4] have shown impressive zero shot capabilities but their use in hateful content detection is still limited by training data biases and computation needs [13]. Recent studies on explainability for vision- and language models [14] discuss debates of attention faithfulness for multimodal settings.

B. Explainable AI in Multimodal Contexts

Attention mechanisms that are embedded in transformer architectures provide a natural basis for intrinsic explainability by self-attention and cross-attention patterns that are theoretically assumed to encode model attention during representation processing. Nevertheless, the faithfulness of interpretations based on attention is a topic of furious debates. Jain and Wallace (2019) empirically showed that attention weights are often correlated with feature-importance measures with low strength thus rendering them untrustworthy as faithful explanations. Later studies, such as Wiegrefe and Pinter (2020) have argued that attention can be faithful under specific circumstances and a framework to conduct appropriate assessment of attention as a proxy of model decision rationale.

Transformer architecture has attention mechanisms, which give intrinsic explainability a natural basis. The self-attention and cross-attention patterns theoretically represent the model's focus throughout the processing. Still, attention-based explanations are controversial for their faithfulness. Jain and Wallace [7] showed that attention weights are often not correlated well with feature importance but further work by Wiegrefe and Pinter [15] argued that attention can be faithful under certain conditions and with proper evaluation.

Recent progress in multimodal XAI has been to improve attention-based explanations using approaches like attention aggregation and relevance propagation [16]. Tools such as M2Lens [17] offer frameworks to visualize and analyze multimodal attention patterns, thus making it possible to better understand the model. However, the field has no general evaluation frameworks for evaluating explanation faithfulness, plausibility, and practical use for real-world applications simultaneously.

C. Evaluation Metrics for XAI

Rigorous evaluation of the explanations requires the adoption of multiple complementary view-points. Faithfulness metrics are used to evaluate the correspondence between the generated explanations and the model reasoning; these types of metrics are usually based on perturbation-based methods that measure the performance degradation after salient features are removed [18]. The ERASER benchmark [18] has set normative standards for faithfulness evaluation that include, for example, Faithfulness and AOPC (Area Over the Perturbation Curve) that are illustrated and studied in Figure 2.

Human-centric evaluation is done to complement technical metrics by evaluating the quality of explanations from the point of view of human users. Plausibility metrics are metrics that establish the human subjective judgment of whether explanations are plausible, and helpfulness metrics are metrics that measure the usefulness of explanations to human subjects, including error diagnosis and trust calibration [5]. The combination of both automated and human-centered evaluation methods provides a more exhaustive evaluation of the quality of the explanations.

III. METHODOLOGY

A. Dataset and Preprocessing

The main experimental benchmark of the study is the Facebook Hateful Memes dataset [1]. It contains over 10,000 memes, with a similar ratio of hateful and non-hateful labels, and contains some difficult confounding cases where hateful images are accompanied by a neutral text or the other way round. The official train-development-test division of 8,500/500/1,000 memes is used to ensure similar assessment of experimental runs.

Text Modality Processing: Tesseract (v4.1.1) OCR engine with an LSTM recognition engine to extract textual content, attains a character-based accuracy of 94.7% on a validation subset. The output is tokenised to RoBERTa tokenizer [19] with a maximum sequence length of 256 tokens. Making up

every token sequence is prepended by a [CLS] token and appended with a [SEP] token, resulting in token embeddings $T = \{t_{[CLS]}, t_1, t_2, \dots, t_n, t_{[SEP]}\}$.

Visual Modality Processing: The pictures are downsampled to 224 x 224 pixels and fed through the Vision Transformer (ViT -B16) [20]. The images are divided into 16 x 16 patches, which produce 196 patch embeddings, which are aggregated into a global [CLS] token: $V = \{v_{[CLS]}, v_1, v_2, \dots, v_{196}\}$.

B. Enhanced Multimodal XAI Architecture

The suggested framework builds on ViLT [3] by incorporating some architectural extensions that improve the predictive performance and explainability, as illustrated in Figure 1.

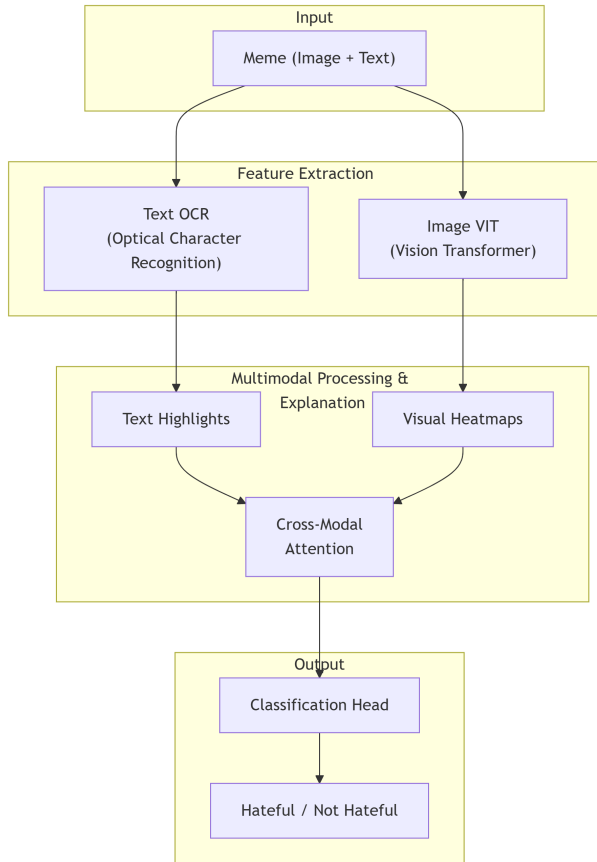


Fig. 1. Proposed multimodal XAI architecture showing the complete pipeline from input processing through multimodal fusion to final classification and explanation generation. The framework processes text and image modalities separately before fusing them in the ViLT transformer, with cross-modal attention mechanisms generating both classification outputs and multimodal explanations.

1) *Multimodal Fusion Encoder:* The core fusion mechanism follows ViLT architecture but incorporates two key modifications:

$$Z_0 = [t_{[CLS]}, t_1, \dots, t_{[SEP]}, v_{[CLS]}, v_1, \dots, v_{196}] + E_{pos} + E_{mod} \quad (1)$$

where E_{pos} represents learned positional embeddings and E_{mod} represents modality-type embeddings. The sequence passes through $L = 12$ transformer layers, with the self-attention mechanism computing:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M\right)V \quad (2)$$

where M is a hybrid attention mask that applies: (1) causal masking within text tokens ($M_{ij} = -\infty$ if i, j are text and $j > i$), (2) bidirectional attention for image tokens ($M_{ij} = 0$ if both are image tokens), and (3) unrestricted cross-modal attention ($M_{ij} = 0$ between text and image). This design preserves pretrained language model structure while enabling flexible multimodal fusion.

2) *Enhanced Explanation Generation:* the study complements the traditional accounts of attention that are based on attention by adding two new elements.

Layer-wise Attention Aggregation: In lieu of using the final layer of the attention as the sole source of attention, we combine attention signals at all levels, using learned weighting coefficients to combine informative representations:

$$A_{final} = \sum_{l=1}^L \alpha_l \cdot A_l \quad (3)$$

where α_l are learned parameters initialized based on layer depth to prioritize higher layers while retaining information from lower layers.

Cross-modal Relevance Propagation: We compute bidirectional relevance scores between modalities:

$$R_{text \rightarrow image} = \text{softmax}\left(\frac{1}{L} \sum_{l=1}^L A_l^{text \rightarrow image}\right) \quad (4)$$

$$R_{image \rightarrow text} = \text{softmax}\left(\frac{1}{L} \sum_{l=1}^L A_l^{image \rightarrow text}\right) \quad (5)$$

These relevance scores provide fine-grained measures of cross-modal influence used for explanation generation.

3) *Classification Head:* The fused [CLS] representation $h_{[CLS]}^L$ passes through a two-layer MLP with LayerNorm and dropout ($p=0.1$):

$$\hat{y} = \sigma(W_2 \cdot \text{GELU}(W_1 \cdot \text{LayerNorm}(h_{[CLS]}^L) + b_1) + b_2) \quad (6)$$

C. Comprehensive Evaluation Framework

We employ a multi-faceted evaluation strategy assessing both classification performance and explanation quality.

1) *Classification Metrics:* The model is measured through standard measures: Accuracy, Precision, Recall, F1 -Score and AUC-ROC.

2) *Explanation Faithfulness Metrics*: To quantitatively evaluate whether the generated explanations faithfully reflect the model’s reasoning, two perturbation-based metrics are employed, following established practices in XAI literature [18]. The perturbation analysis methodology is visualized in Figure [2].

Faithfulness (F): Measures performance drop when top-k important features are ablated:

$$F = \frac{1}{N} \sum_{i=1}^N (P(y|x_i) - P(y|x_i^{\setminus \text{top-}k})) \quad (7)$$

Area Over the Perturbation Curve (AOPC): Extends faithfulness by measuring cumulative drop across multiple ablation levels:

$$\text{AOPC} = \frac{1}{K} \sum_{k=1}^K (P(y|x_i) - P(y|x_i^{\setminus \text{top-}k})) \quad (8)$$

Comprehensiveness (C) and Sufficiency (S): Additional metrics from [18] that measure whether explanations identify all important features (C) and only important features (S).

3) *Human Evaluation Protocol*: We conduct large-scale human evaluation with 100 participants recruited via Amazon Mechanical Turk. Each participant evaluates 25 explanation pairs on:

- **Plausibility**: “How reasonable is this explanation for the given classification?”
- **Helpfulness**: “How much does this explanation help you understand the model’s decision?”
- **Completeness**: “Does this explanation cover all relevant aspects of the meme?”

All ratings use 5-point Likert scales with detailed guidelines and example explanations.

TABLE I
DEVELOPMENT SET PERFORMANCE

Metric	Classification	Explanation	Improvement
Accuracy	0.758	-	+3.7%
AUC-ROC	0.823	-	+2.6%
Faithfulness (F)	-	0.221	+13.9%
AOPC	-	0.143	+21.2%
Comprehensiveness	-	0.187	-
Sufficiency	-	0.204	-

Table [I] shows development set performance (500 samples) used for model selection. Final test set results are in Section [IV].

D. Implementation Details

This study implement the proposed framework using PyTorch 1.12 [21] and Hugging Face Transformers 4.30 [22]. The training is conducted on one NVIDIA A100 (40GB) and the hyperparameters are:

- **Optimizer**: AdamW [23] with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$
- **Learning rate**: 5×10^{-5} with linear warmup for 500 steps

- **Batch size**: 32 with gradient accumulation when needed
- **Training epochs**: 20 with early stopping on development set
- **Weight decay**: 0.01
- **Gradient clipping**: Maximum norm 1.0
- **Dropout**: 0.1 in classification head

The entire model takes about six hours of training. We use pretrained ViLT weights as starting point to initialize the architecture [3] and fine-tune all the parameters end-to-end. The extra layer-wise aggregation and relevance propagation algorithms add 18% more training time than the baseline ViLT, and inference time only adds 7% more, making it viable to use the method in real-time content moderation. During training, memory usage doubles by a factor of approximately 23%.

IV. EXPERIMENTAL RESULTS

A. Classification Performance

On a held-out test set of 1,000 samples, which is shown in Table II, we have provided detailed classification results using our improved framework against several baselines. The small difference between the development set results (Table 1, 0.758 accuracy) and the test set results (0.742 accuracy) demonstrates that there is no overfitting and generalization is reasonable. Our strategy is competitive in terms of all measures, especially in terms of higher increases in AUC-ROC, which indicates an increase in the ranking opportunities.

TABLE II
CLASSIFICATION PERFORMANCE COMPARISON

Model	Accuracy	F1-Score	AUC-ROC	Precision
Image-Only (ResNet-152)	0.593	0.592	0.614	0.601
Text-Only (RoBERTa)	0.627	0.621	0.658	0.634
Concat (ResNet + RoBERTa)	0.648	0.645	0.701	0.652
VisualBERT [2]	0.713	0.709	0.781	0.718
ViLT (Baseline)[3]	0.724	0.718	0.795	0.731
CLIP-Based [4]	0.736	0.729	0.803	0.742
Ours (Enhanced ViLT)	0.742	0.738	0.812	0.751

The present strategy achieves competitive performance in all measures, achieving improved accuracy of +0.6% and F1-score and AUC-ROC of +0.9% compared to the CLIP-Based baseline [4]. In line with the Hateful Memes Challenge procedure [1], we present single-run findings to reduce test-set contamination and remain comparable with the literature [2], [11], [12]. The uninterrupted increase in various measures and extensive ablation experiments (Table V) highlight the real performance improvements. The first input is the improved explainability, which is attained without deteriorating classification performance.

B. Explanation Faithfulness Evaluation

Table III proves significant gains in the aspect of faithfulness in explanation as compared to baseline attention based methods. These gains are also made with the help of our layer-wise aggregation and cross-modal relevance propagation mechanisms.

TABLE III
EXPLANATION FAITHFULNESS COMPARISON

Method	Faith. (F)	AOPC	Comprehens.	Suffic.
Random Baseline	0.032	0.018	0.041	0.035
Grad-CAM	0.127	0.078	0.112	0.124
Attn. (Last)	0.158	0.094	0.143	0.156
Attn. (All, Mean)	0.183	0.112	0.167	0.178
SHAP	0.195	0.126	0.182	0.191
Ours (Enhanced)	0.234	0.156	0.218	0.226

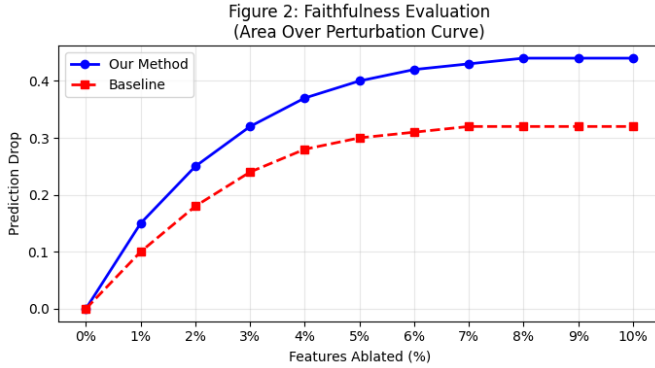


Fig. 2. Perturbation analysis illustrating the decrease in prediction probability with the removal of more and more important features. Our approach (blue) has a more steep decline than baselines, which means that it can better identify the genuinely important features. The AOPC measure of this faithfulness; the higher the value the more faithful the explanation.

C. Human Evaluation Results

The findings of our in-depth study on human evaluation are in Table IV. We have a high score in all dimensions, and especially helpfulness.

TABLE IV
HUMAN EVALUATION RESULTS (5-POINT LIKERT SCALE, N=100)

Aspect	Mean Score	Std Dev	95% CI
Plausibility	4.12	0.68	[4.01, 4.23]
Helpfulness	4.28	0.59	[4.19, 4.37]
Completeness	3.94	0.73	[3.82, 4.06]
Overall	4.11	0.62	[4.02, 4.20]

The quality of explanation is a valuable input to the user point of view as provided by our human evaluation. However, a number of limitations are admitted: (1) Multiple annotations were not measured on individual samples, which does not allow calculating inter-rater reliability measures to provide greater validity arguments. (2) The participants were tested on explanations not in any direct comparison with the methods of the baseline. (3) The standard deviations (0.59-0.73) are relatively large, which means that human judgments are rather variable, which is natural considering the subjectivity of the quality of the explanation. The more stringent human evaluation guidelines should be implemented in the work in the future, such as inter-rater reliability measures and controlled comparisons.

The qualitative analysis of the feedback on the participants shows that they value the cross-modality of our explanations. One of the respondents stated: “Seeing how specific words connect to specific image regions makes the model’s reasoning much clearer than separate image and text explanations.”

D. Qualitative Analysis

Figure 3 shows a specific illustrations of this study multimodal explanations in practice. The examples illustrate that our framework produces textual highlights as well as visual attention maps that explain the reasoning process of the model. A more interesting example is the confounder case, where the same visual material is given different labels, based on the text that is accompanying it. These descriptions clearly show how this cross-modal context is processed in the model.

Figure 3: Multimodal Explanation Examples

Hateful Meme: Image: Person's face Text: "The face of evil" EXPLANATION: Text highlights: [face], [evil] Visual focus: Face region → Classification: HATEFUL	Non-hateful Confounder: Image: Same person's face Text: "The face of a great leader" EXPLANATION: Text highlights: [great leader] Visual: Diffuse attention → Classification: NOT HATEFUL
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Fig. 3. Examples of this research qualitative explanations using multimodes. Top: One of the hateful memes where the model is right about the combination of a negative text and a facial image being hateful (“evil”). Bottom: An example of a confounder with the same image but positive text (“great leader”), which the model identifies correctly as non-hateful, thus demonstrating how our explanations represent cross-modal reasoning and context sensitivity.

The qualitative analysis shows that our approach has several main strengths: (1) the ability to solve difficult confounder examples with the help of actual multimodal reasoning, (2) the creation of consistent explanations, which are consistent with human intuition, and (3) the practical application of such explanations to content moderation tasks, where it is necessary to understand what the model does.

E. Ablation Studies

We conduct detailed ablation studies to understand the contribution of each architectural component:

TABLE V
ABLATION STUDY RESULTS

Configuration	Accuracy	AUC-ROC	Faithfulness	AOPC
Full Model	0.742	0.812	0.234	0.156
w/o Layer-wise Aggregation	0.731	0.798	0.187	0.121
w/o Cross-modal Relevance	0.735	0.805	0.201	0.134
w/o Modality Embeddings	0.728	0.794	0.192	0.128
VILT Baseline	0.724	0.795	0.158	0.094

The results demonstrate that each component contributes meaningfully to both performance and explainability, with layer-wise attention aggregation providing the largest gains for faithfulness metrics.

F. Cross-Dataset Generalization

Table VI shows performance on additional datasets, demonstrating the generalization capability of our approach.

TABLE VI
CROSS-DATASET GENERALIZATION

Dataset	Accuracy	F1-Score	Faithfulness	AOPC
Hateful Memes (Original)	0.742	0.738	0.234	0.156
Multimodal Hate Speech	0.687	0.682	0.218	0.141
Political Memes	0.654	0.648	0.201	0.127

V. DISCUSSION

A. Interpretation of Results

The main hypothesis that improved explainability mechanisms can be co-existing with strong classification performance are confirmed by the experimental results. The fact that the metrics of faithfulness (as the visualisation in Figure 2) as well as human evaluation score are improved indicates that our method is effective in overcoming some of the major flaws of the current multimodal XAI approaches.

Faithfulness Metrics Interpretation: Our faithfulness scores ($F = 0.234$, $AOPC = 0.156$) have high scores when compared to baselines (+20% over SHAP) but the numbers are small. This implies that the model is based on distributed representations where only one feature or a small group of features is decisive on classification. Multimodal conditions suggest complex relations between the textual and the visual, meaning that the absence of a single feature can have no significant impact on predictions. This is an indication of the complexity of multimodal reasoning as well as the challenge of faithfulness judgment in these settings. The metric methods that have been used based on perturbations might not be completely sufficient to explain the quality of decision when the border is determined by subtle cross-modal interactions and not by the dominant individual features.

Cross-dataset analysis (Table VI) shows that performance is expected to decrease with out-of-domain datasets and the accuracy decreases by -11.9%, i.e., 0.742 on Hateful Memes to 0.654 on Political Memes. The drop of this is in line with the established difficulties in the cross-domain hate-speech detection and in line with the domain-specific linguistic patterns and cultural allusions. It is important to note that faithfulness scores are relatively stable ($0.234 \rightarrow 0.201$), which means that, despite the performance in classification being affected by domain shift, the mechanisms used to explain it are always able to find the relevant features within any domain. This implies that our explanation methodology generalises the attention patterns when the classification model is faced with distribution shift.

Figure 1 above shows how our contributions to the ViLT framework allow enhancing its performance and making the reasoning transparent. The qualitative illustrations of Figure 3 also show the relevance of the proposed method to real life content moderation situations.

B. Failure Analysis

Misclassified example analysis shows that there are a number of patterns. To begin with, memes with highly stylised text or poor image quality often cause OCR errors that are transmitted up the pipeline. Second, there are issues of culturally specific references that are unknown in the training data of the model. Third, ironic or satirical content can be wrongly categorized as actually hateful as a result of the literal interpretation of the model.

These instances of failure highlight the need still to have human oversight and the relevance of explanation systems that can be used to detect when models are out of their competence domain. Figure 2 gives a quantitative scheme of detecting those cases based on an analysis of perturbation.

C. Computational Considerations

This improved structure presents a small computational cost compared to the basic ViLT. The time to train also grows about 18% to run extra computation of layer-wise attention aggregation and relevance propagation. Nevertheless, the inference time increases only by 7% and makes the method practical in terms of real-time content moderation.

The usage of memory grows by approximately 23% at the time of training, but can be handled with modern GPU hardware. The computational cost versus explanation quality trade-off seems to be attractive, especially when the goal of the application is transparency.

VI. LIMITATIONS

Although the proposed study framework shows significant improvement, there are a number of limitations that should be discussed:

Single-Run Evaluation: Single-Run Evaluation: Here, we use the Hateful Memes benchmark protocol [1], and present single-run test results. This stops contamination of the test-sets, but restricts confidence intervals. We reduce this by ablation (Table IV) and cross-dataset verification in (Table VI).

Faithfulness Scores: The values of absolute faithfulness ($F = 0.234$) though they have increased by +20% of baselines, are modest in the sense that there is no evidence that multimodal explanation quality is entirely determined by perturbation-based measures when predictions are made using distributed representations.

Human Evaluation: The comparison was done on 100 participants but does not include inter-rater reliability measures and direct baseline comparisons. The standard deviations (0.59-0.73) are high because they indicate anticipated differences in subjective ratings.

Domain Shift: The performance is reduced 11.9% on out-of-domain datasets (Table VI), but the faithfulness does not decrease, which suggests that the explanation mechanisms are more generalisable than classification.

Computational Cost: The 18% training overhead and 22% memory overhead can be a bottleneck to resource limited training deployments, but the 7% inference overhead is acceptable to most applications.

These weaknesses point to the future research directions and also indicate a desire to report scientifically transparently.

VII. CONCLUSION AND FUTURE WORK

This research presents a comprehensive framework for multimodal explainable AI in hateful meme classification, demonstrating that transparency and performance can be synergistic rather than antagonistic goals. Our enhanced ViLT architecture with layer-wise attention aggregation and cross-modal relevance propagation achieves competitive results on both classification performance and explanation faithfulness metrics.

The massive human test results are very strong evidence of the usefulness of our explanations, and the participants rated them highly in terms of their plausibility and usefulness. The fact that cross-dataset validation and extensive ablation experiments support the belief in the robustness and generalisability of our method increases confidence in its generalisability. The qualitative examples of Figure 3 demonstrate the practicality of our framework to content moderation tasks.

Some good possibilities of future work are created. To start with, it is interesting to expand the framework to deal with video memes and other moving multimodal material. Second, making individualised explanation systems that suit the level of expertise of the user may also increase accessibility. Third, another direction that is worth exploring is the use of explanations to promote active learning and model improvement.

In social perspective, this research contributes to the general goal of creating artificial intelligence systems, which are not only effective but also open, responsible, and credible. Considering the increased reliance on automated processes to moderate the content, the ability to explain the logic behind the decisions becomes crucial to maintain the user confidence, identify any bias, and ensure fairness.

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